

HANG VICTOR

Site Reliability Engineer



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EXPERIENCES

Senior Tech Lead - Platform Team | StackHPC

Since Apr, 2026

Joined the Platform team as a Senior Tech Lead, working on Kubernetes and OpenStack infrastructure.

Site Reliability Engineer | S3NS

Jun, 2023 - Mar, 2026

As a Site Reliability Engineer on the Compute & Virtualization team, I am responsible for engineering the core of a sovereign cloud platform from a Thales-Google Cloud partnership. My primary mission is to design, automate, and operate the internal Infrastructure-as-a-Service (IaaS) layer, ensuring high availability, robust security, and strict compliance with the French government's SecNumCloud standards.

DevOps Consultant at Orange Business Service | SFEIR

Nov, 2022 - Jun, 2023

As a Level 3 DevOps Consultant for SFEIR, I was embedded with the client, Orange Business Services, to serve as a senior technical authority. My role involved architecting resilient cloud-native solutions, resolving complex systemic issues, and mentoring client teams on DevOps best practices for automation, security, and large-scale observability.

DevOps Administrator | SysOps | CHEOPS TECHNOLOGY

Jun, 2021 - Oct, 2022

As a key member of a dedicated DevOps squad, I was tasked with the design and construction of an internal Kubernetes-as-a-Service (KaaS) platform. The goal was to deliver a fully self-service, automated, and secure platform for internal development teams to build, deploy, and manage their applications.

EDUCATION

DevOps Consultant | ib - groupe Cegos

2021

Network and System Administration | EPSI

2020

CERTIFICATIONS

GCP Architect

CKA Kubernetes Administrator

Gitlab Associate

PUBLICATIONS

PROJECTS

Home Cluster

A comprehensive, personal Kubernetes-as-a-Service (KaaS) platform designed to replicate a professional cloud-native environment for home use. Built on Proxmox for virtualization, the entire infrastructure is declaratively managed with NixOS and a GitOps workflow powered by FluxCD. This ensures perfect reproducibility and automated configuration management. The architecture achieves secure multi-tenancy by using Kamaji to run Kubernetes control planes as pods within a management cluster, providing strong isolation. Identity and access are centralized through Zitadel, while HashiCorp Vault manages secrets securely. The platform serves as a powerful environment for hosting personal media servers, home automation, and for experimenting with advanced cloud-native technologies.

Netwatch

A powerful security and operations tool for Kubernetes that automates the management of temporary network access. It addresses the common operational need for granting developers or SREs short-term access to specific services (e.g., a database for debugging) without leaving permanent security holes. Netwatch introduces a 'NetworkAccess' Custom Resource (CRD) where a user can request access with a specified duration. The Netwatch controller then automatically generates a corresponding, temporary Kubernetes NetworkPolicy. Once the defined duration expires, the controller automatically removes the NetworkPolicy, ensuring the principle of least privilege is maintained and the cluster remains secure.

MaxTac

A Kubernetes controller created to abstract away the complexity of writing and managing native Kubernetes NetworkPolicy resources. It solves the problem of NetworkPolicies being verbose, difficult to reason about, and prone to misconfiguration. MaxTac introduces two intuitive Custom Resource Definitions (CRDs)—'Access' for internal traffic and 'ExternalAccess' for ingress/egress. These CRDs provide a simpler, higher-level abstraction for developers to define connectivity rules based on application logic rather than low-level pod and port selectors. The controller translates these simple CRDs into the appropriate, more complex NetworkPolicy YAML, reducing cognitive load and improving the security posture of the cluster.

Songbird

A command-line interface (CLI) tool that acts as a linter and simulator for Kubernetes network connectivity, designed to prevent network-related outages. In a complex cluster with numerous NetworkPolicies, it's challenging to predict the outcome of a change. Songbird solves this by allowing operators to proactively evaluate their NetworkPolicy configuration. It can simulate traffic flow between any two points in the cluster and report whether the current policy set would allow or deny the connection. This enables teams to validate their network security rules in a CI/CD pipeline or locally before applying them to a live cluster, thereby preventing accidental service disruptions.

NixBook

A declarative and modular NixOS configuration framework specifically tailored for laptops. It addresses the classic 'it works on my machine' problem by allowing a developer's entire operating system—from the kernel and system packages to application configurations and dotfiles—to be defined as code using Nix Flakes. This ensures a perfectly reproducible, version-controlled environment that can be easily shared and deployed across multiple machines. The framework is designed with customizable profiles and modules, making it simple to manage different setups (e.g., for work and personal use) from a single, clean codebase.

NixOS-Server

A server-focused NixOS configuration that applies declarative and GitOps principles to manage bare-metal infrastructure. This repository contains a complete, code-defined setup for servers that host Kubernetes clusters. Every aspect of the server environment, including networking, storage, system packages, and security hardening, is defined in Nix code and managed in a Git repository. Changes to the infrastructure are made via a git push, which triggers an automated deployment process. This provides a powerful audit trail, enables atomic updates and rollbacks, and ensures that the entire server fleet remains in a consistent, predictable, and reproducible state.

SKILLS

Cloud & Infrastructure, Development & Automation, Systems & Databases

LANGUAGES

French (Native) , English (Professional)

HOBBIES

Coding, Gym, Concerts, Travels